

Diagnosing Reversed and Non-Equivalent Ratios

Find the mistake, name it, and fix it: reversed ratios, broken tables, and false equivalences

Directions:

- Each problem shows work that might contain a ratio mistake. Decide what (if anything) went wrong, then fix it.
- Two classic errors to hunt for: writing the numbers in the **reversed order**, and **adding** to both parts of a ratio instead of multiplying.

1. A recipe uses 3 cups of oats and 5 cups of milk. Kim writes the ratio of **milk to oats** as 3:5. Write the correct ratio of milk to oats in the blank, and say what Kim got mixed up.
correct ratio of milk to oats: _____

2. Jon says the ratios 4:6 and 10:15 are equivalent. Is he right? Write = or $\neq$ in the blank, and explain how you know by simplifying both ratios in the work space.

$$\frac{4}{6} \quad \text{_____} \quad \frac{10}{15}$$

3. A car travels 120 miles in 3 hours. To find its speed, Leo computes $3 \div 120 = 0.025$ and reports “0.025 miles per hour.” Find the correct speed, and name the mistake Leo made.

Answer: _____ mph

4. The table is supposed to show rice and milk in the ratio 2:3, but one entry breaks the pattern.

cups of rice	2	4	6	8
cups of milk	3	6	10	12

- (a) Circle the entry that does not belong.
- (b) What should that entry be? _____

5. A batch of 8 muffins needs 2 cups of flour. To find the flour for 12 muffins, Mia writes the proportion

$$\frac{2}{8} = \frac{12}{x}$$

and gets $x = 48$ cups. Explain the error in Mia's proportion, then find the correct number of cups for 12 muffins.

6. Three students each claim their ratio is equivalent to 3:4. The claims are 6:8, 9:10, and 15:20 — but one claim is false.

- (a) Which ratio is **not** equivalent to 3:4? _____
- (b) Fix it: keep its first number and write what the second number should be. _____

7. Punch A uses 3 cups of syrup for every 7 cups of water. Punch B uses 4 cups of syrup for every 9 cups of water. Raj writes $\frac{7}{3} > \frac{4}{9}$ and concludes, “Punch A is much stronger.” In the work space, say what Raj mixed up. Then compare the punches fairly: write the correct comparison of syrup-per-water fractions using $<$ or $>$.

$$\frac{3}{7} \quad \text{—————} \quad \frac{4}{9}$$

8. The table is supposed to show the ratio 5:2, but **two** of the bottom entries are wrong.

top	10	15	20	25
bottom	4	8	8	12

- (a) Corrected bottom entry under 15: _____
- (b) Corrected bottom entry under 25: _____
- (c) Explain the test you used on each column to decide whether it belongs.